# FX Correlation-Regime Forecasting

An edge-centric graph transformer with a GRU temporal head, built to test whether graph structure over a cross-asset market graph predicts shifts in FX pair correlation beyond what simple linear mechanics already capture. The target is the forward change in trailing 20-day correlation, `corr[t+H] - corr[t]`, for EUR-GBP, EUR-JPY, and GBP-JPY.

## Result

At the 5-day horizon, the graph model:

- outperforms a no-graph plain-GRU control on mean IC (0.288 vs 0.270; margin +0.019, 95% CI [−0.044, +0.076], P(margin>0) = 0.733),
- beats a 2-state Gaussian HMM regime baseline (0.288 vs −0.082),
- but only ties — while still trailing — a calibrated linear mean-reversion baseline (0.288 vs 0.329; margin −0.046, 95% CI [−0.113, +0.012]).

The graph-vs-GRU margin leans positive in 73% of block-bootstrap resamples, but overlapping estimation windows and forecast horizons leave only ~24 effectively independent test blocks — too few to certify an edge this small at the 95% level. The uplift from relational context is directionally suggestive but statistically unresolved; a longer test period, not a different architecture, is what would settle it.

At the 20-day horizon the graph loses to the linear baseline outright. So across both horizons it never beats linear mechanics — it ties at best.

Conclusion: the predictable structure in these correlation shifts is largely linear and mean-reverting; the graph architecture adds a small, statistically unresolved edge over a plain GRU and no advantage over a two-feature linear model.

## Why the evaluation is built this way

Early experiments scored deceptively well until a trivial baseline matched the model — the target was mechanically predictable (mean-reverting), so high IC reflected mechanics, not skill. The evaluation was redesigned around this: a calibrated linear baseline defines the mechanical bar, a no-graph ablation isolates the architecture's contribution, and permutation/bootstrap tests gate every claim. The goal is to measure skill *beyond* mechanics, and to report a negative result rather than a flattering one.

## Method

- **Spatial:** a per-day edge transformer over a 25-node graph (3 FX pairs + 22 macro context nodes — rates, yields, equities, volatility, commodities),

learning representations on the 6 directed target edges. Backpropagation updates edge representations directly rather than deriving edges from node embeddings — chosen because the objective is local relational forecasting in a noisy domain.
- **Temporal:** a GRU over 20-day windows of edge states predicting the forward correlation shift.
- **Evaluation:** time-purged train/val/test splits, train-only standardization, a calibrated linear baseline and no-graph ablation, with significance via block-permutation tests and bootstrap confidence intervals.

## Repo layout

- `src/models/` — graph transformer, edge GRU
- `src/training/train_graph.py` — end-to-end training
- `src/evaluation/` — evaluate, significance tests, HMM baseline, context-signal gate
- `runs/` — run logs (run005–run007)
- `docs/Engineering_log.md` — dated decision log

## Data

Daily 25-node macro-FX graph, 1999–2026 (not committed — large binaries, gitignored). Place `X.npy`, `nan_mask.npy`, `dates.csv`, `node_names.csv` in `data_live_2026/`. Sourced from FRED (set `FRED_API_KEY` in `.env`).

## Running

```
pip install -r requirements.txt
python -m src.training.train_graph                    # trains ->
checkpoints/best_model.pt
$env:FX_HORIZON=5; python -m src.evaluation.evaluate   # scores + significance
```

## Limitations & future work

- Single seed; seed-robustness not yet run.
- Both neural models dip sharply in 2024Q2 where the linear baseline holds — a regime-dependent weakness, not yet diagnosed.
- HMM baseline is a standard 2-state Gaussian; a Student-t or higher-state variant is the natural extension, not yet explored.

## Reproducibility

- Data schema: `X.npy` is `[T, 25, 3]` — per node [daily return, level z-score, vol]; node order in `node_names.csv`; `nan_mask.npy` marks missing cells (True = masked). Dates in `dates.csv`.
- Splits: train ≤ 2021-12-31, validation ≤ 2023-12-31, test 2024-01-02 → 2026-05-21 (599 predictions). Standardization is fit on train only.

- Config via env vars: `FX_HORIZON` (default 20), `FX_SEED` (0), `FX_CORR_W` (20), `FX_SEQ_LEN` (20), `FX_GRU_HIDDEN` (24). All reported results use seed 0 and horizon 5 unless noted.